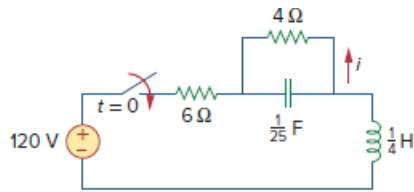


Problem

In the circuit of Fig. 16.74, find $i(t)$ for $t > 0$.

Figure 16.74:



Step-by-step solution

Step 1 of 8

Refer to circuit diagram in Figure 16.74 in the textbook.

Switch is open for a long time before switch is opened at $t = 0$. For $t < 0$, inductor is short circuit and capacitor is open circuit. The equivalent circuit is shown in Figure 1.

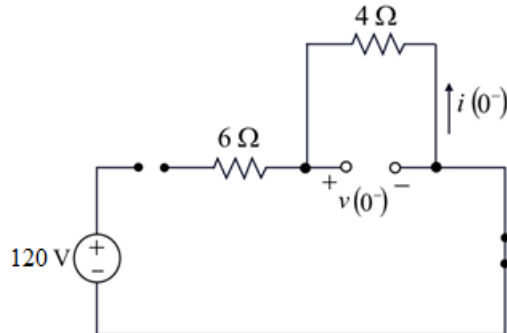


Figure 1

Step 2 of 8

The value of current $i(0^-)$ is,

$$i(0^-) = 0 \text{ A}$$

Inductor doesn't allow sudden changes in the current flowing through it.

$$\begin{aligned} i(0^-) &= i(0) \\ &= i(0^+) \\ &= 0 \text{ A} \end{aligned}$$

The value of voltage $v(0^-)$ is,

$$v(0^-) = 0 \text{ V}$$

Capacitor doesn't allow sudden changes in the voltage across it.

$$\begin{aligned} v(0^-) &= v(0) \\ &= v(0^+) \\ &= 0 \text{ V} \end{aligned}$$

Step 3 of 8

For $t > 0$, switch is closed.

The s-domain circuit for $t > 0$ is shown in Figure 2.

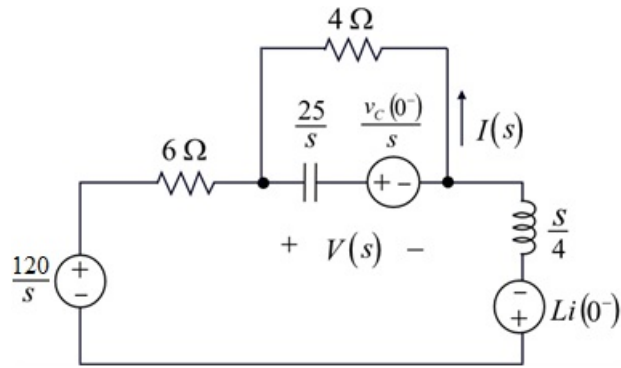


Figure 2

Step 4 of 8

Modified circuit diagram of Figure 2 is as shown in Figure 3.

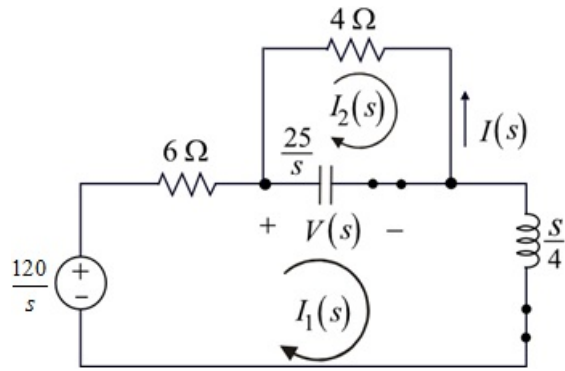


Figure 3

Step 5 of 8

Apply Kirchhoff's voltage law to Loop 1 in Figure 3.

$$-\frac{120}{s} + 6I_1(s) + \frac{25}{s}[I_1(s) - I_2(s)] + \frac{s}{4}I_1(s) = 0$$

$$\left(6 + \frac{25}{s} + \frac{s}{4}\right)I_1(s) - \frac{25}{s}I_2(s) = \frac{120}{s}$$

$$I_1(s) = \frac{\frac{25I_2(s) + 120}{s}}{\left(\frac{24s + 100 + s^2}{4s}\right)}$$

$$= \frac{100I_2(s) + 480}{s^2 + 24s + 100}$$

Step 6 of 8

Apply Kirchhoff's voltage law to Loop 2 in Figure 3.

$$4I_2(s) + \frac{25}{s}[I_2(s) - I_1(s)] = 0$$

$$\left(4 + \frac{25}{s}\right)I_2(s) - \frac{25}{s}I_1(s) = 0$$

Substitute $\frac{100I_2(s) + 480}{s^2 + 24s + 100}$ for $I_1(s)$.

$$\left(4 + \frac{25}{s}\right)I_2(s) - \frac{25}{s}\left(\frac{100I_2(s) + 480}{s^2 + 24s + 100}\right) = 0$$

$$\left(\frac{4s + 25}{s}\right)I_2(s) - \frac{2500I_2(s)}{s(s^2 + 24s + 100)} = \frac{25(480)}{s(s^2 + 24s + 100)}$$

$$\left(\frac{(4s + 25)(s^2 + 24s + 100) - 2500}{s(s^2 + 24s + 100)}\right)I_2(s) = \frac{25(480)}{s(s^2 + 24s + 100)}$$

$$I_2(s) = \frac{25(480)}{4s^3 + 121s^2 + 1000s + 2500 - 2500}$$

$$= \frac{3000}{s(s^2 + 30.25s + 250)}$$

Step 7 of 8

The current $I(s)$ is,

$$I(s) = -I_2(s)$$

$$= -\frac{3000}{s(s^2 + 30.25s + 250)}$$

Apply partial fractions.

$$I(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 30.25s + 250} \dots\dots (1)$$

$$-\frac{3000}{s(s^2 + 30.25s + 250)} = \frac{A(s^2 + 30.25s + 250) + Bs^2 + Cs}{s(s^2 + 30.25s + 250)}$$

$$-3000 = A(s^2 + 30.25s + 250) + Bs^2 + Cs \dots\dots (2)$$

Calculate the value of A .

Compare the constant terms on both sides in equation (2).

$$250A = -3000$$

$$A = -12$$

Calculate the value of B .

Compare the s^2 coefficient terms on both sides in equation (2).

$$A + B = 0$$

$$-12 + B = 0 \quad (\text{Since } A = -12)$$

$$B = 12$$

Calculate the value of C .

Compare the s coefficient terms on both sides in equation (3).

$$30.25A + C = 0$$

$$(30.25)(-12) + C = 0 \quad (\text{Since } A = -12)$$

$$C = 363$$

Step 8 of 8

Recall the equation (1).

$$I(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 30.25s + 250}$$

Substitute -12 for A , 12 for B and 363 for C .

$$\begin{aligned} I(s) &= \frac{-12}{s} + \frac{12s + 363}{s^2 + 30.25s + 250} \\ &= \frac{-12}{s} + \frac{12s + 363}{(s + 15.125)^2 + (4.608)^2} \\ &= \frac{-12}{s} + \frac{12(s + 15.125) + 181.5}{(s + 15.125)^2 + (4.608)^2} \\ &= \frac{-12}{s} + \frac{12(s + 15.125)}{(s + 15.125)^2 + (4.608)^2} + \frac{39.39(4.608)}{(s + 15.125)^2 + (4.608)^2} \end{aligned}$$

Apply inverse Laplace transform to the equation.

$$\begin{aligned} i(t) &= -12u(t) + 12e^{-15.125t} \cos(4.608t)u(t) + 39.39e^{-15.125t} \sin(4.608t)u(t) \\ &= \left\{ -12u(t) + (41.17)(0.2914)e^{-15.125t} \cos(4.608t)u(t) + \right. \\ &\quad \left. (41.17)(0.9565)e^{-15.125t} \sin(4.608t)u(t) \right\} \\ &= -12 + (41.17)e^{-15.125t} \left[(0.2914) \cos(4.608t) + (0.9565) \sin(4.608t) \right] u(t) \\ &= -12 + (41.17)e^{-15.125t} \left[\frac{\cos(73.06) \cos(4.608t) + \sin(73.06) \sin(4.608t)}{\sin(73.06)} \right] u(t) \\ &= -12 + (41.17)e^{-15.125t} \cos(4.608t - 73.06^\circ) u(t) \text{ A} \end{aligned}$$

Thus, the expression for current through the resistor $i(t)$ for $t > 0$ is

$$\boxed{-12 + (41.17)e^{-15.125t} \cos(4.608t - 73.06^\circ) u(t) \text{ A}}.$$